DRAFT — every red [TBD] must be filled from experiment outputs before submission

BrainHybridNet: A Lightweight Hybrid CNN–Transformer for Joint Brain Tumor Segmentation, Detection and Grading with Self-Supervised Pre-training and Uncertainty Estimation

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***Abstract*—Objective: Automated brain tumor analysis from magnetic resonance images usually requires separate models for segmentation, detection and grading, and most report no measure of confidence. We present a single lightweight model that performs all three tasks and estimates its own uncertainty. Methods: The proposed network combines a two-stage convolutional stem, which preserves high-resolution detail, with a four-layer transformer that models long-range volumetric context. In one forward pass it segments the whole tumor, tumor core and enhancing tumor, detects tumor presence and predicts low- versus high-grade glioma. The encoder is pre-trained by reconstructing masked regions of unlabeled scans; repeated stochastic inference with dropout yields a voxel-wise uncertainty map, and transformer attention provides spatial explanations. Five established networks were re-trained on identical patient-level splits of the 2021 Brain Tumor Segmentation challenge data (1,251 patients) for comparison. Results: On the held-out test set the model reaches Dice scores of [TBD], [TBD] and [TBD] for whole tumor, tumor core and enhancing tumor, with [TBD] million parameters and [TBD] milliseconds per volume crop; [TBD: grading area under the curve, calibration error and external-validation result]. Conclusion: [TBD: measured trade-off between accuracy, model size and calibration]. Significance: A compact model that segments, detects and grades brain tumors while flagging uncertain regions can support radiologist review on standard clinical hardware.**

***Index Terms*—Brain tumor segmentation, glioma grading, magnetic resonance imaging, masked image modeling, transformer, uncertainty estimation.**

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I. INTRODUCTION

Glioblastoma (WHO grade 4) has a median overall survival of 14.6 months under standard treatment [1], and about 308,000 new brain and central nervous system tumors were estimated worldwide in 2020 [2]. Delineating the whole tumor (WT), tumor core (TC) and enhancing tumor (ET) on multi-modal MRI (T1, contrast-enhanced T1ce, T2, FLAIR) [3] guides surgery, radiotherapy and response assessment, but manual delineation is slow and raters agree only to Dice 0.74–0.85 [4].

CNNs such as 3D U-Net [5, 6] capture local structure but not long-range context; transformers [7, 8] do, at high data and compute cost. Hybrid designs [9–11] combine both, but most solve one task, give point predictions only [12] and have tens of millions of parameters [13].

We propose BrainHybridNet. Our contributions are:

- a lightweight hybrid encoder (CNN stem with high-resolution skips, four-layer transformer bottleneck) with **[TBD]** million parameters;
- 3D masked image modeling applied before the CNN stem of a hybrid encoder, extending ViT self pre-training on target data [14], with measured effects on accuracy and label efficiency;
- joint segmentation, detection and LGG/HGG grading in one pass;
- Monte Carlo dropout uncertainty, evaluated for calibration and correlation with error, and attention maps evaluated against reference masks;
- a reproducible comparison with five re-trained baselines on identical splits, with confidence intervals, paired tests and external validation; code, splits and weights are public.

II. RELATED WORK

Segmentation networks. 3D U-Net [6], SegResNet [15] and the self-configuring nnU-Net [16] are strong CNN baselines. UNETR [17] and SwinUNETR [18, 19] use transformer encoders; TransBTS [10], MedT [20], UCTransNet [21] and D-Former [11] mix convolution and attention.

Self-supervised learning. Contrastive learning [22–24] and masked image modeling [25, 26] reduce label needs; MAE self pre-training improves BraTS segmentation [14].

Uncertainty. Bayesian networks [27], deep ensembles [28] and Monte Carlo dropout [12] estimate epistemic uncertainty; in neuroimaging it has been linked to errors [29] and used for quality control [30].

Grading. WHO CNS5 grading uses histology and molecular markers [31]; imaging-based grading uses radiomics [32] or CNNs [33], usually separate from segmentation.

III. METHOD

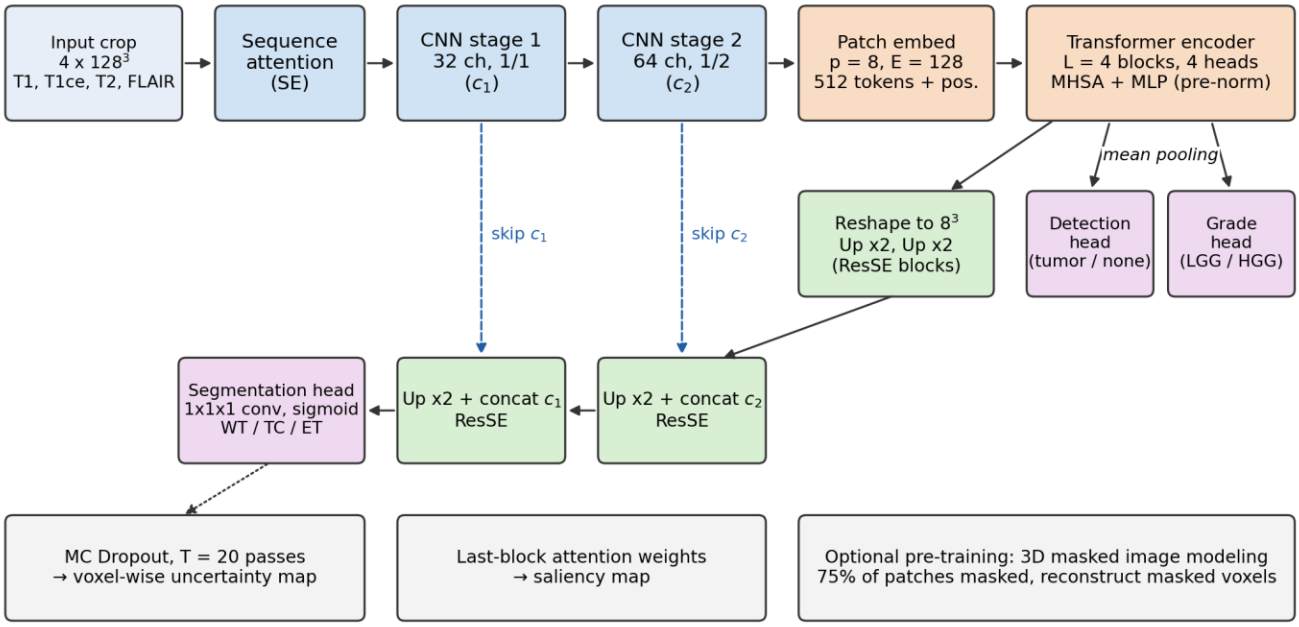


Fig. 1. BrainHybridNet. Sequence attention and a two-stage CNN stem give skip features c_1, c_2; 512 tokens from 8^3 patches of c_2 pass four transformer blocks. Pooled tokens feed detection and grade heads; the decoder fuses c_2 and c_1 before the WT/TC/ET head. Monte Carlo dropout gives uncertainty, last-block attention a saliency map.

A. Architecture

Fig. 1 shows the network. Input $\mathbf{X} \in \mathbb{R}^{B\times4\times D\times H\times W}$ (four z-scored sequences, 128^3 crops) is re-weighted per sequence by squeeze-and-excitation and passed through two stem stages: $c_1 = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3,1}(\mathbf{X})))$ (32 channels, full resolution) and $c_2 = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3,2}(c_1)))$ (64 channels, half resolution). A strided convolution turns 8^3 patches of c_2 into $N = 512$ tokens of size $E = 128$ with learned positional embeddings $\mathbf{P}$, followed by $L = 4$ pre-norm transformer blocks (4 heads, MLP ratio 4, GELU [34], dropout 0.1) [35]:

$$z'_l = z_l + \mathrm{MHSA}(\mathrm{LN}(z_l)),\quad z_{l+1} = z'_l + \mathrm{MLP}(\mathrm{LN}(z'_l)) \qquad (1)$$

The decoder reshapes tokens to an 8^3 grid, applies 3D dropout (0.1), and upsamples four times with transposed convolutions and residual squeeze-and-excitation (ResSE) blocks, concatenating c_2 and c_1 at matching resolutions. Heads: a 1×1×1 convolution with sigmoids for WT/TC/ET; detection Linear(E,128) → ReLU → Linear(128,2) and grading Linear(E,64) → ReLU → Linear(64,2) on mean-pooled tokens. As BraTS has no tumor-free subjects, detection is trained per crop (positive if ≥100 WT voxels); subjects without grade labels are ignored by the grading loss.

B. Masked image modeling

Following SimMIM [26] in 3D, 75% of token positions are masked; their 16^3 input regions are zeroed before the stem (so nothing leaks through convolutions) and the tokens replaced by a learned mask embedding. A linear head reconstructs the 4×16^3 voxels of each masked region; the loss is the mean squared error to per-patch normalized targets over masked patches only. Pre-training uses only the training folds.

C. Uncertainty

With batch normalization frozen and dropout active [12], $T = 20$ sliding-window passes give the mean μ and voxel-wise variance $\sigma^2 = (1/T)\,\Sigma_t\,(\sigma(\hat{Y}^{(t)}) - \mu)^2$.

D. Training

$\mathscr{L} = 0.6\,\mathscr{L}_{\mathrm{seg}} + 0.2\,\mathscr{L}_{\mathrm{det}} + 0.2\,\mathscr{L}_{\mathrm{grade}}$, with cross-entropy for detection and grading and the Tversky loss [36] ($\alpha = 0.3$, $\beta = 0.7$) per region for segmentation. AdamW [37] (learning rate 10^{-4}, weight decay 10^{-4}, clipping 1.0), 10 warm-up epochs then cosine decay [38], 300 epochs (pre-training 100 epochs at 1.5×10^{-4}), batch 2, mixed precision; flips, intensity scaling/shift and Gaussian noise; crops centered on tumor with probability 0.66; best mean validation Dice per fold.

IV. EXPERIMENTAL SETUP

Data. BraTS 2021 [39]: 1,251 subjects, four co-registered skull-stripped sequences at 1 mm^3; WT, TC and ET from the expert labels. A subject-level hold-out test set (15%, **[TBD]** subjects) is used once; the rest forms five folds shared by all models. Grading uses UCSF-PDGM [40] (WHO grade), which also serves as an external segmentation test set.

Baselines. 3D U-Net, SegResNet, UNETR, SwinUNETR (MONAI [41]) and nnU-Net v2, all re-trained on the same folds and preprocessing.

Metrics. Dice and HD95 per region (BraTS conventions for empty masks); AUC, balanced accuracy, sensitivity and specificity; voxel-level expected calibration error [42]; Spearman correlation of case uncertainty with 1 − Dice; parameters, MACs, latency and memory on **[TBD: GPU]**. Mean ± SD, 95% bootstrap intervals, Wilcoxon tests with

Holm correction. Reporting follows CLAIM [43]. Implementation: PyTorch [44], PyTorch Lightning [45].

V. RESULTS

TABLE I
SEGMENTATION ON THE BRATS 2021 HOLD-OUT TEST SET ([TBD] SUBJECTS), MEAN ± SD

Method	Dice ↑			HD95 (mm) ↓			Params (M)	GMACs	Latency (ms)
	WT	TC	ET	WT	TC	ET			
3D U-Net [6]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
SegResNet [15]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
UNETR [17]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
SwinUNETR [18]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
nnU-Net v2 [16]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
BrainHybridNet	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

TABLE II
DETECTION (BRATS CROPS) AND GRADING (UCSF-PDGM)

	Det. AUC	Sens.	Spec.	Grade AUC	Grade BA
BrainHybridNet	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

Segmentation. Table I; significance versus the strongest baseline: **[TBD]**. Five-fold cross-validation: **[TBD]**.

Detection and grading. Table II.

Calibration and uncertainty. **[TBD: ECE with/without Monte Carlo dropout; Spearman ρ and p]**.

External validation. **[TBD: UCSF-PDGM without retraining]**.

TABLE III
ABLATIONS (5-FOLD CROSS-VALIDATION, DICE)

Configuration	WT	TC	ET	Params
CNN only	[TBD]	[TBD]	[TBD]	[TBD]
No CNN skips	[TBD]	[TBD]	[TBD]	[TBD]
No sequence attention	[TBD]	[TBD]	[TBD]	[TBD]
Segmentation only	[TBD]	[TBD]	[TBD]	[TBD]
No pre-training	[TBD]	[TBD]	[TBD]	[TBD]
Full model	[TBD]	[TBD]	[TBD]	[TBD]

Ablations. Table III; masking ratio, depth and loss-weight sensitivity: **[TBD]**.

VI. DISCUSSION

[TBD: interpret only the measured results: where the hybrid helps, the effect of pre-training with 10%/25% labels, whether uncertainty tracks error, attention-map overlap].

Limitations. External validation is limited to **[TBD]**; grading ignores molecular markers [31]; pre-training uses only BraTS training folds; detection is evaluated on crops, not healthy scans; reference labels vary between raters [4].

VII. CONCLUSION

BrainHybridNet jointly segments, detects and grades brain tumors with a compact hybrid encoder, masked-image-modeling pre-training and uncertainty estimates. **[TBD: one-sentence measured summary]**. Code, splits and weights: https://github.com/Debobrato06/BrainTumorDetection **[TBD: release tag]**.

ACKNOWLEDGMENT

[TBD: or remove].

DATA AVAILABILITY

BraTS 2021 (https://www.synapse.org/#!Synapse:syn27046444) and UCSF-PDGM (The Cancer Imaging Archive) are public; this study used only these de-identified datasets, so no ethics approval was required.

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